DATA UNDERSTANDING AND PREPARATION PACKAGE

Week 1 Deliverable

Prepared by Group 12

DATA DICTIONARY

The Data Dictionary is a structured spreadsheet that explains the variables present in the Opportunity dataset. It provides a clear reference for understanding and consistently using the dataset during analysis.

The spreadsheet contains the following key information for each variable:

1. Column Name – the name of the variable in the dataset.
2. Description – an explanation of what the column represents.
3. Data Type – the type of data contained in the column, such as Text, Number, Date/Time, or Boolean.
4. Important Notes/Observations – relevant information about the variable, including data-quality issues, standardizations, missing values, or other observations.

Data Dictionary Spreadsheet:

Data Dictionary (sheet 2)

1. Overview of the Dataset

The source workbook, contains 5,733 Opportunity records and 33 columns on the source sheet “Sheet1.” The fields include identifiers, opportunity classifications, dates, fees, duration information, application settings, location, descriptive text, and several relationship/metadata fields stored as JSON-like text.

After preparation, the final analytical dataset contains 5,733 records and 33 columns. No duplicate opportunity IDs were identified, and no records were removed because of duplicate IDs.

2. Data Quality Issues Identified

- Mixed formatting in categorical fields, including variations such as Virtual/virtual/vitrual and Work From Home/WFM/wfm.
- Singular and plural duration units were mixed (for example, week/weeks and month/months), with a real count of clearly invalid entries.
- Currency values included both EUR and EURO; EURO was standardized to EUR.
- Boolean columns contained TRUE/FALSE values alongside a real count of malformed text entries.
- Date fields were stored as Unix epoch milliseconds rather than analysis-friendly date/time values.
- Some fields contained literal placeholders such as Null or {} instead of blank/missing values.

- A real count of cells contained clearly corrupted or misplaced text in fields where the expected data type was known.
- Several descriptive text fields contained inconsistent whitespace.
- The dataset contains substantial legitimate missingness in optional fields; these values were not artificially filled.
- Several relationship fields contain nested JSON-like structures. These were retained as text because flattening them without business rules could change their meaning.

3. Cleaning Steps Performed

1. Trimmed leading/trailing whitespace from text fields.
2. Converted explicit blank/Null/{} placeholders to missing values where they represented absent information.
3. Converted fee, duration, and microscholarship to numeric data types.
4. Converted is_archived and is_auto_approve to TRUE/FALSE boolean values; malformed entries were set to missing.
5. Converted created_at, modified_at, and last_date_to_apply from Unix epoch milliseconds to Excel-compatible date/time values. An invalid zero application deadline was treated as missing.
6. Standardized duration units to weeks, months, hours, days, minutes, or years.
7. Standardized currency EURO to EUR.
8. Standardized location values to consistent labels, including Virtual and Work From Home; obvious placeholders and unusable one-off entries were set to missing.

9. Retained only recognized category values and set clearly corrupted one-off category entries to missing.
10. Validated image_link values at a basic format level by retaining HTTP/HTTPS URLs and setting non-URL corrupted values to missing.
11. Normalized whitespace in long_description, short_description, summary, and role_responsibility without removing substantive wording.
12. Checked for duplicate rows and duplicate opportunity IDs. No duplicate opportunity IDs were removed.
13. Did not impute business values such as fees, scholarships, categories, or descriptions.

4. Assumptions and Decisions

- Missing values were preferred over guessing when a corrupted value could not be reliably reconstructed from the dataset.
- The source system's nested relationship/JSON fields were treated as text because their relational structure was outside the scope of this cleaning exercise.
- The opportunity_id field was treated as the record-level unique identifier.
- The existing numeric values for fee, duration, and microscholarship were preserved; no outlier was removed solely because it appeared unusually large or small.
- Future application deadlines were retained because they may represent valid upcoming opportunities.

- No business logic was applied to determine whether an opportunity itself was valid, active, or desirable; the cleaning focused on data consistency and analytical usability.

5. Final Dataset Summary

Metric	Result
Rows before cleaning	5,733
Rows after cleaning	5,733
Columns	33
Duplicate opportunity IDs removed	0
Date fields converted	3
Numeric fields standardized	3
Boolean fields standardized	2
No business-value imputation	Yes

The cleaned workbook is suitable as the Week 2 starting point for exploratory analysis and dashboard development. The remaining missing values should be interpreted as genuine unknown/optional values unless a later business rule or source-system reference can provide a reliable replacement.

CLEANED DATASET

The Cleaned Dataset is the prepared version of the Opportunity dataset, reviewed and organized to make it suitable for Week 2 exploratory analysis and dashboard development.

The dataset addresses major data-quality issues identified during the cleaning process, including inconsistent formatting, data types, missing or invalid values, and other inconsistencies where applicable. The cleaned dataset has been organized to ensure that the information is clear, consistent, and easy to understand.

The cleaning applied to the dataset aligns with the cleaning steps documented in the Data Cleaning Documentation, providing a reliable and analysis-ready dataset for further exploration and visualization.

Cleaned Dataset Spreadsheet:

Cleaning Summary (Sheet 3)

DELIVERABLES

The accompanying Excel workbook contains three sheets: (1) Cleaned Opportunity, the analysis-ready dataset; (2) Data Dictionary, describing every field, data type, and important observations; and (3) Cleaning Summary, documenting the main transformations and before/after missing-value counts.